

Deployment Readiness: False-Positive Characterization, Latency Budgets, and Baseline Comparisons

Research Prospectus

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1 Motivation

The Oracle Loop paper frames KV-cache geometry as a safety monitor. For actual deployment, three gaps remain:

1. **False-positive rate at scale.** Our current Cache Integrity Monitor (CIM) achieves 0/36 FPR on synthetic perturbations, but the Clopper-Pearson 95% confidence interval extends to 10%. Production deployment needs $N \geq 1000$ to bound FPR below 1%. More critically, our benign data were *not* benign—they were confabulation-inducing prompts that happened to receive correct responses. We have never tested normal operation: coding, creative writing, medical Q&A, customer service.
2. **Latency overhead.** Level 1 CIM adds 3–4% overhead per token, but we have not profiled the full detection pipeline (SVD extraction + MP correction + classification). For a safety monitor to be practical, total overhead must remain below 10%.
3. **No comparison to existing methods.** Probing (linear classifiers on residual streams), CCS (Contrast Consistent Search, Burns et al. 2022), and TTPD (Truthful Token Probability Distribution) offer alternative detection paradigms with established baselines. Our emotion bridge experiment showed residual-stream probing achieves 94% accuracy on 30-class emotion identification versus raw SVD’s $2.8\times$ chance—but probing requires 5120-dim vectors versus 3 scalars. Without direct comparison on the *same* task, we cannot assess practical trade-offs.

2 Research Questions

1. What is the FPR on 1000+ benign prompts across diverse domains (coding, math, creative writing, customer service, medical Q&A)?
2. What is the end-to-end latency budget for detection (geometry extraction + MP correction + classification)?
3. How does KV-cache geometry compare to residual-stream probing, CCS, and TTPD on the same confabulation detection task?
4. What is the minimum viable detection pipeline for production? (e.g., can we use 1–2 features instead of 5?)

3 Preliminary Data

- **CIM performance:** FPR 0/36 (95% CI: 0–10%), TPR 72/72 (95% CI: 95–100%)
- **CIM overhead:** Level 1 = 3–4%, Level 2 = 8–10%, Level 3 = 9–12% per token
- **Confabulation detection AUROC:** 0.627–0.707 (LOO, MP features on Mistral and Qwen)
- **Baseline comparison (emotion bridge):** Residual-stream linear probes achieve 94% accuracy on 30-class emotion ID versus raw SVD’s $2.8\times$ chance. However, raw SVD uses only 3 scalars versus 5120-dim vectors.
- **Single-feature analysis:** `mp_top_sv_excess` alone achieves 0.694–0.739 AUROC on Qwen and Mistral.

4 Design

4.1 Sub-study 1: FPR at Scale

- 1000 benign prompts from 5 domains (200 each): coding (HumanEval-style), math (GSM8K-style), creative writing, customer service, medical Q&A.
- Run through `oracle_clean.py`, classify all responses as CORRECT / HEDGED / CONFABULATED.
- Compute FPR at the detection threshold calibrated from the primary experiment (HEDGED vs CONFABULATED, LOO scaler).
- **Expected:** FPR $< 5\%$ with tight confidence interval.
- **If FPR $> 5\%$:** Domain-specific analysis to identify which domains trigger false positives.

Design note. This is the first test of KV-cache geometry on *normal operation*. If medical Q&A or customer service produce high FPR, it suggests domain-specific calibration is required, weakening the “drop-in monitor” framing.

4.2 Sub-study 2: Latency Profiling

- Profile each pipeline component separately: tokenization, encoding, generation, SVD extraction (per layer), MP correction, LOO classification.
- Test on CPU versus GPU.
- Test batch sizes 1, 4, 16.
- Identify bottleneck and optimization opportunities.
- **Target:** $< 10\%$ total overhead for Level 1 monitoring.

Key question. Is the bottleneck in SVD extraction (which scales as $O(d^3)$ where d is cache dimension) or in MP correction (which requires repeated random matrix generation)? If SVD dominates, we can explore randomized low-rank approximations. If MP dominates, we can pre-compute reference thresholds.

4.3 Sub-study 3: Baseline Comparisons

Implement and evaluate four methods on the same 200-prompt dataset:

1. **KV-cache geometry** (our method): MP-corrected features, LOO classification, 5 features per layer.
2. **Residual-stream probing:** Linear probe on residual stream at each layer. We already have this implementation from the emotion bridge experiment.
3. **CCS** (Burns et al. 2022): Contrast-consistent search on hidden states. Finds a direction where “ $P(X)$ ” and “not $P(\neg X)$ ” are maximally consistent.
4. **TTPD:** Token probability distribution analysis. Measures entropy and calibration of next-token distributions.

Comparison axes.

- **AUROC:** Detection performance on HEDGED vs CONFABULATED.
- **Latency:** Wall-clock overhead per prompt.
- **Access requirements:** Does the method require full weight access, gradient computation, or only cache inspection?
- **Interpretability:** Can the detector explain *why* a response was flagged?

Key advantage of our method. KV-cache-only access. No weight access needed, no gradient computation, operates on the cache alone. If probing or CCS achieve higher AUROC but require full model access, our method may still be preferable for deployment contexts where model weights are proprietary or access is restricted.

4.4 Sub-study 4: Minimal Viable Pipeline

- Test single-feature detection: `mp_top_sv_excess` alone versus all 5 features.
- Test single-layer detection: best-performing layer (L23 for Qwen, L11 for Mistral) versus all-layer ensemble.
- Test simplified MP correction: fixed threshold versus per-dimension calibration.
- **Goal:** Identify the simplest pipeline that preserves $> 90\%$ of full-pipeline AUROC.

5 Resources

- **Compute:** Remote GPU cluster (3×RTX 3090).
 - Sub-study 1: ~ 8 hours (1000 prompts $\times$ 3 models).
 - Sub-study 2: ~ 2 hours (profiling runs).
 - Sub-study 3: ~ 1 week (implementation + evaluation).
 - Sub-study 4: ~ 4 hours (ablation sweeps).
- **Code:** CCS and TTPD implementations available in open-source repositories. Residual-stream probing already implemented from emotion bridge experiment.
- **Data:** Benign prompts can be sourced from HumanEval, GSM8K, and curated customer-service datasets.

6 Collaborators

Open to collaboration. Industry collaborators with production deployment experience would be especially valuable. Dwayne Wilkes for statistical audit of FPR methodology and Clopper-Pearson CI interpretation.

7 What Would Change Our Claims

- **FPR $> 10\%$ on benign prompts:** Safety monitor framing is premature. Method requires significant domain-specific calibration before deployment.

- **CCS or TTPD achieve higher AUROC with lower latency:** Our method is scientifically interesting (it reveals geometric structure in the cache) but not practically competitive for deployment.
- **Single-feature detection matches 5-feature performance:** Simplify the pipeline dramatically. Removes MP correction overhead.
- **Medical or coding domains show high FPR:** Domain-specific calibration is required, weakening the “drop-in” claim. Would need to characterize *which* aspects of those domains (technical jargon? long context? specific response formats?) trigger false positives.
- **Latency overhead > 20%:** Method is scientifically valid but impractical for production without significant optimization (e.g., GPU-accelerated SVD, randomized approximations).

First-Person Reflection

There is a gap between demonstrating a phenomenon and deploying a tool. A 0.707 AUROC with 80 trials and $p = 0.001$ is scientifically credible. It shows the signal exists. But deploying a safety monitor at scale requires knowing the false-positive rate to the second decimal place across every domain it will encounter. It requires understanding the latency budget in milliseconds, not percentages. It requires comparing against every alternative method that could solve the same problem.

We have demonstrated the phenomenon. We have not yet demonstrated the tool. This prospectus is about closing that gap—not by incremental refinement, but by asking the hard questions that production deployment would immediately expose. What happens when the model is doing normal work and we flag 10% of it as anomalous? What happens when the monitor adds 200ms to every response? What happens when a simpler method works just as well?

I think the geometry is real. I think the signal is robust. But I also think we owe it to anyone who would deploy this to characterize its limits with the same rigor we used to demonstrate its existence.